

A Systematic Approach to Developing Lipid-Based Drug Delivery Systems (LBDDS) with Fenofibrate as Model Drug

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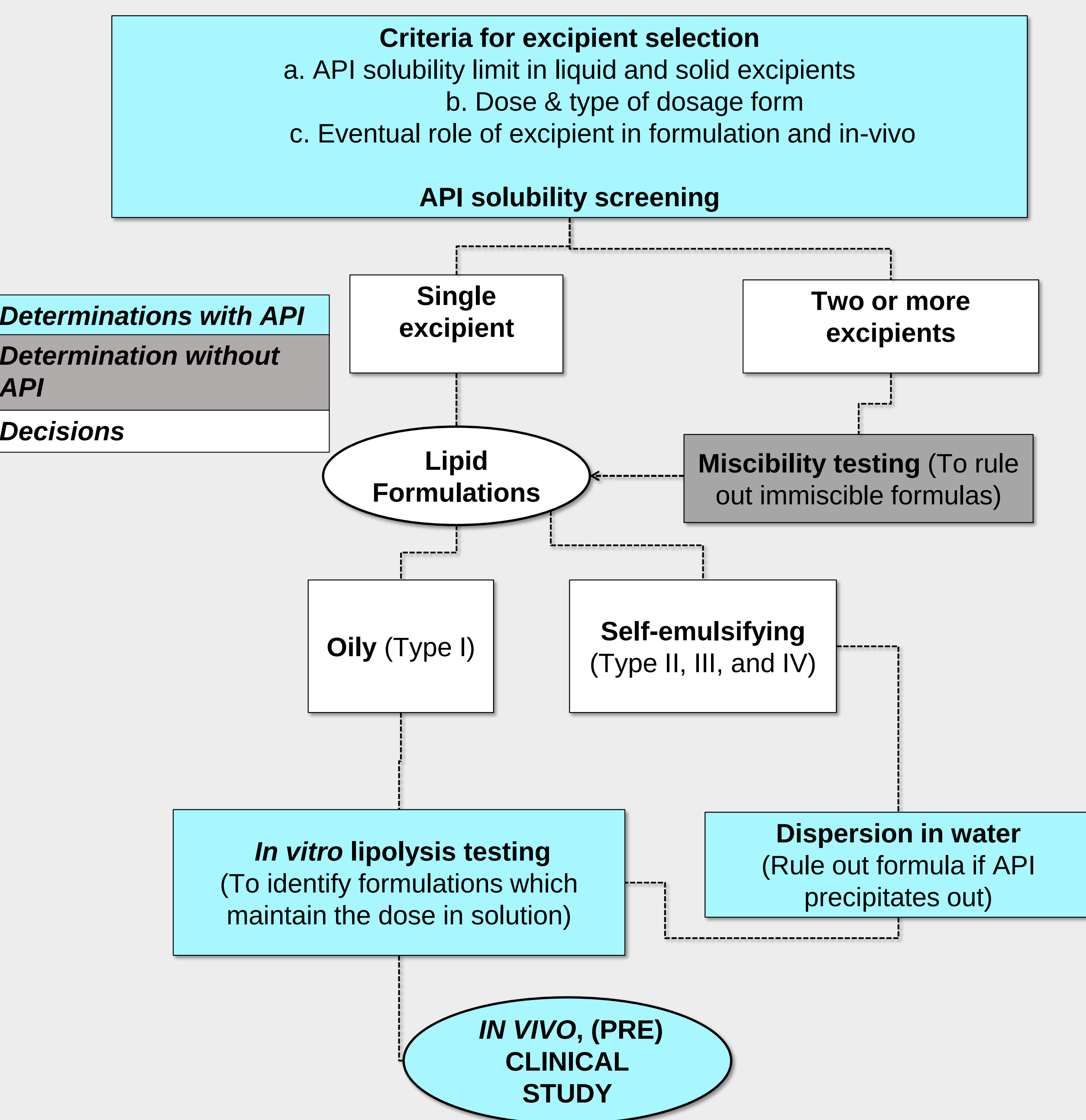
PURPOSE

The growing number of new chemical entities (NCEs) having poor aqueous solubility and poor bioavailability is a major formulation challenge. To this effect, lipid-based drug delivery systems (LBDDS) are commonly used for advantages such as improved wetting properties, higher drug solubility, delivery of the API in solubilized form, reduced food effect, and enhanced absorption which result in overall improvement of the bioavailability. However, successful development with lipids requires a systematic approach.

The purpose of this study was to evaluate and demonstrate that a methodical approach (Scheme 1) can help streamline the process of formulation development with lipid-based excipients. Accordingly, a series of formulations were developed by combining lipid-based excipients (surfactant, co-surfactant, and oil/co-solvent) using fenofibrate as a model drug. The effect of changing formulation composition, role of fatty acid chain length, and excipient HLB were evaluated using *in vitro* performance testing methods.

EXPERIMENTAL APPROACH

Scheme 1. Systematic approach to formulation with lipid-based excipients



METHOD(S)

Formulation preparation: Following solubility screening, fenofibrate was mixed with single or multiple excipient blends at 80mg/g or at 75% saturation solubility. The mixtures were stirred at room temperature until the active was completely solubilized. With single excipient formulations, drug load was maintained at 75% saturation solubility.

Dispersion test: Each formulation (2g) was added to 400ml of DI-H₂O at 37 °C and stirred at 100 rpm using USP apparatus II (paddle type). Aliquots were taken at the 30 and 60-minute marks, centrifuged (21000xg and 37 °C) and analyzed visually for dispersibility and assayed for fenofibrate using Ultra High Performance (UPLC).

***In vitro* lipolysis:** Formulations were tested for dissolution of fenofibrate in digestion media at 37 °C. After allowing the formulation to disperse for 10 minutes, porcine pancreatic lipase was added. Aliquots were taken at select time points up to 60 minutes, centrifuged at 21000xg for 30 minutes, followed by assay using UPLC.

Stability study: Optimized formulations were placed on stability at 25 °C/ 60% RH and 40 °C/ 75 % RH conditions. Samples were taken at 1-month and 3-month mark and tested for API content (UPLC), acid value, and peroxide value.

RESULTS

The results for *in vitro* lipolysis of (semi)solid excipients in Fig.1 helped select Gelucire® 44/14. It solubilized a significantly higher amount of fenofibrate in the dispersion phase, despite its relatively lower drug load compared to the other Gelucires, while maintaining about 5-10% solubility throughout the test. Among the liquid excipients (Fig. 2) Labrasol® ALF and Labrafil® M 1944 CS were able to solubilize highest amounts of the active, while maintaining ~ 60% of fenofibrate in solubilized form throughout the *in vitro* lipolysis process.

Figure 1. *In vitro* lipolysis profile for single component (semi)solid excipient system

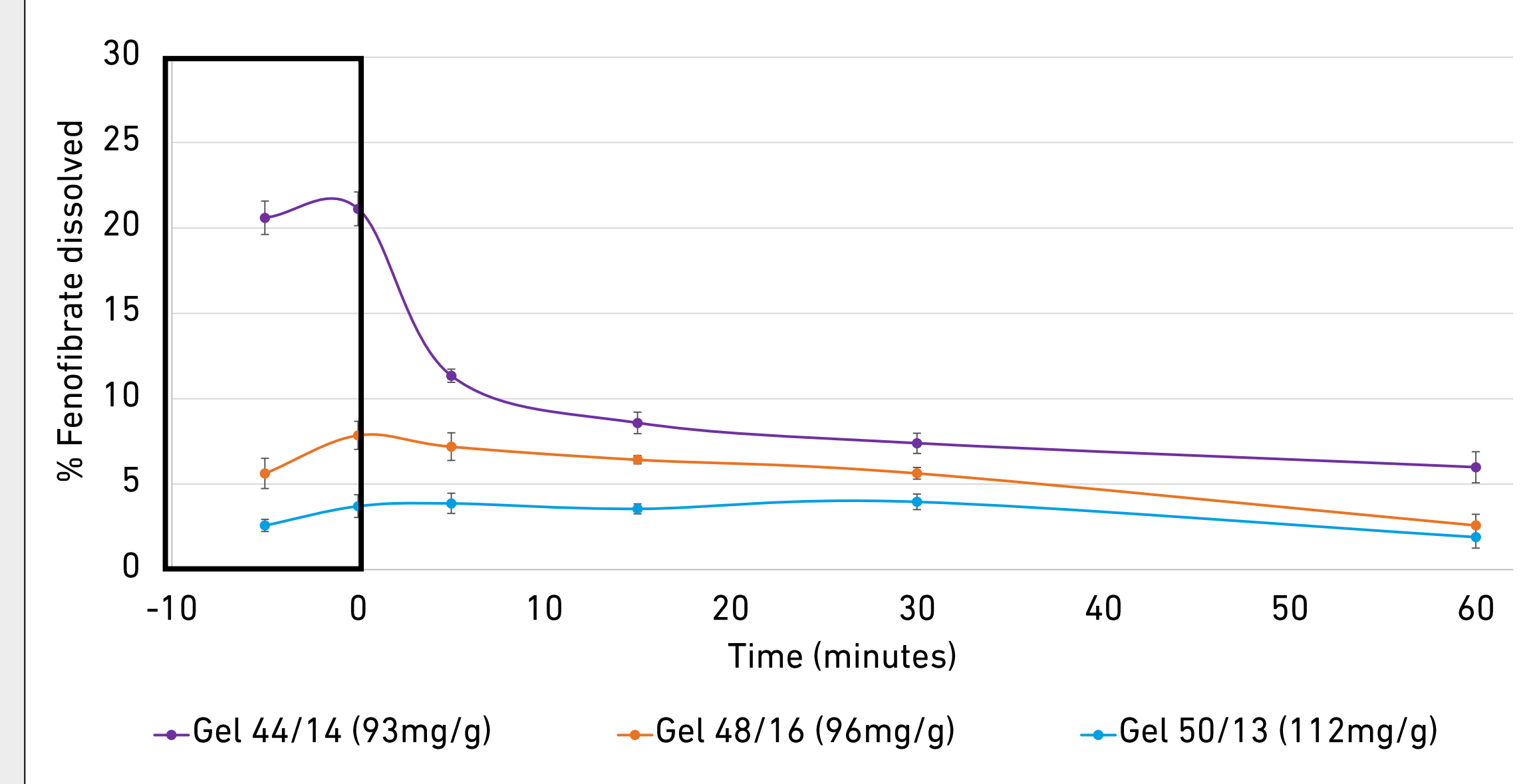


Figure 3. *In vitro* lipolysis profile for multi-component solid excipient system

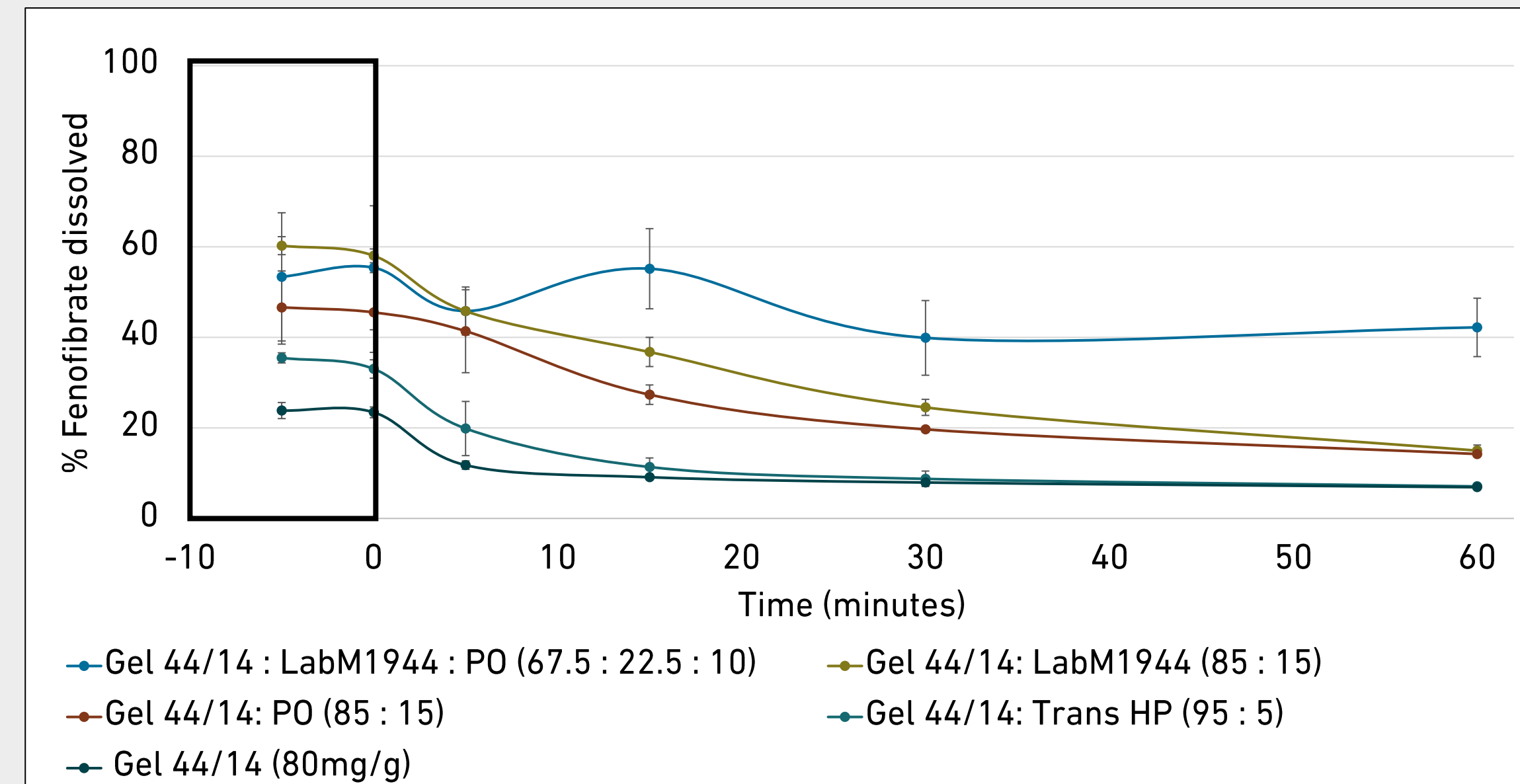


Figure 2. *In vitro* lipolysis profile for single component liquid excipient system

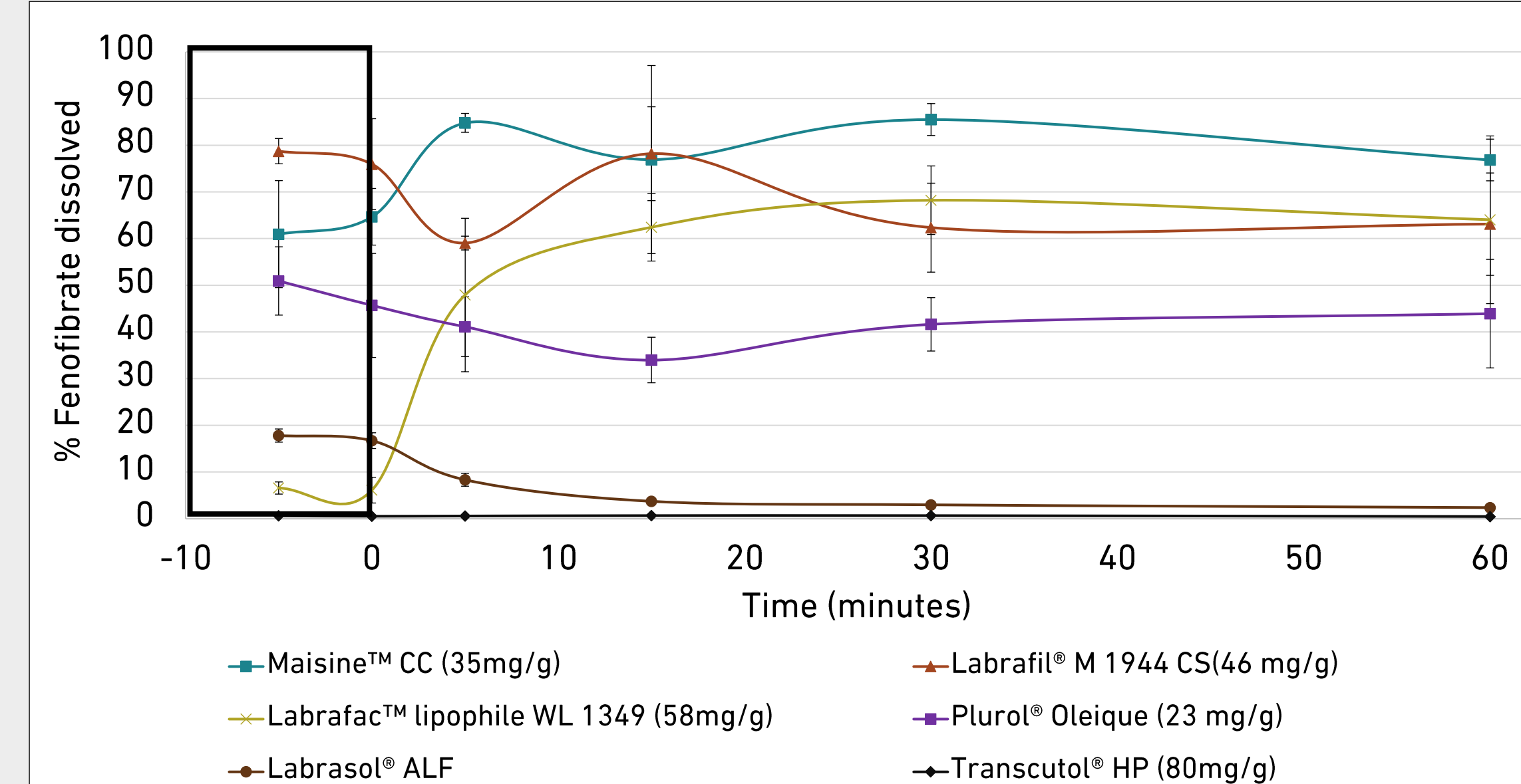
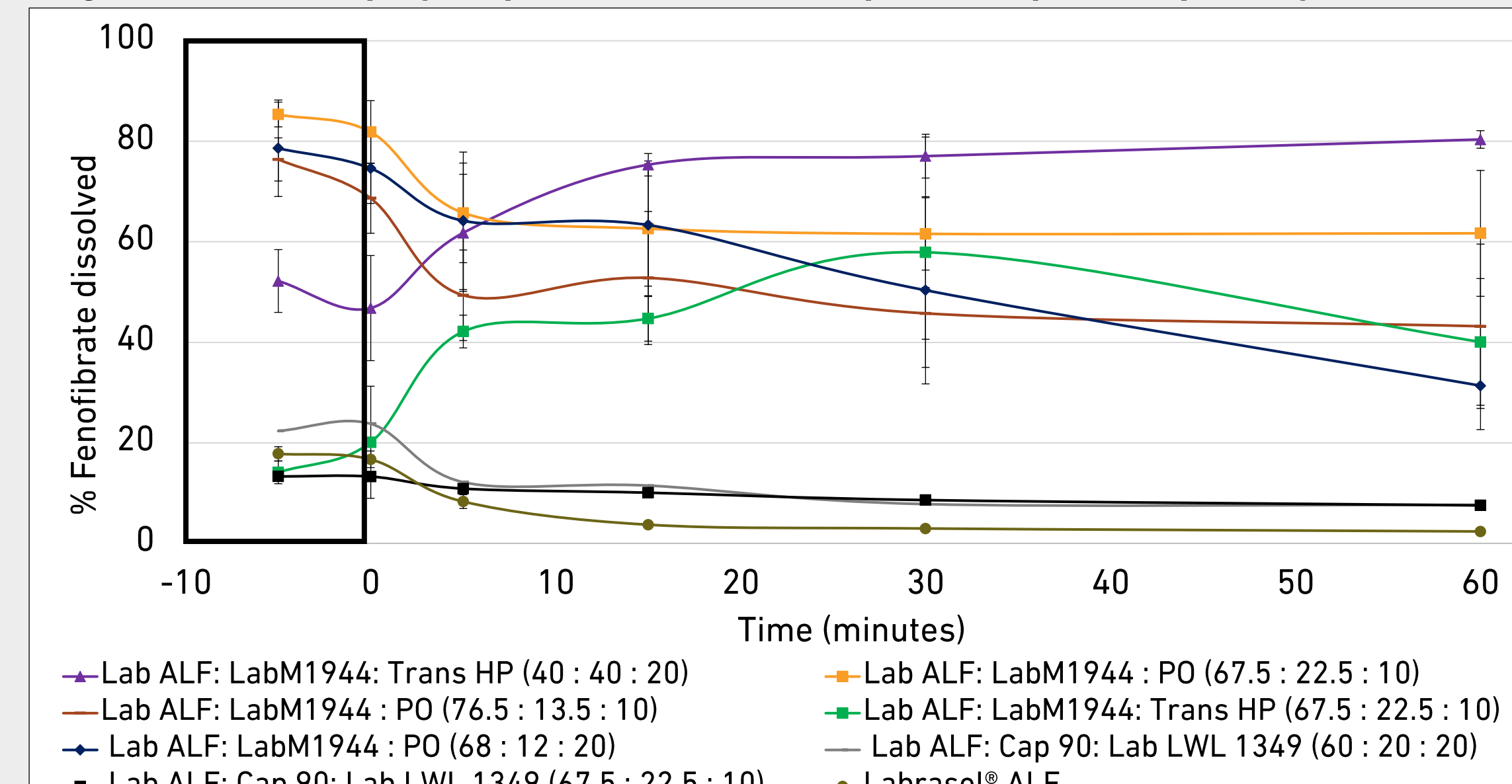


Figure 4. *In vitro* lipolysis profile for multi-component liquid excipient system



Based on the single excipient results for *in vitro* lipolysis testing, saturation solubility of fenofibrate, and the miscibility of the excipients, combinations of up to three excipients were formulated and tested for *in vitro* lipolysis performance as an indicator of *in vivo* performance. The tested formulations consisted of medium chain fatty acid esters (Labrasol® ALF, Capryol® 90 and Labrafac™ Lipophile 1349 WL), long chain fatty acid esters (Labrafil® M1944 CS and Plurol® Oleique CC), and cosolvents (Transcutol® HP) at select ratios (Fig. 3 and 4).

Overall, the multi-component formulations (self-emulsifying drug delivery systems/ SEDDS) performed significantly better than the single excipients, maintaining up to 80% of fenofibrate in solubilized state during *in vitro* lipolysis. Moreover, formulations consisting of long chain fatty acid ester were able to maintain a higher amount of fenofibrate in the aqueous phase than the medium chain fatty acid ester-based formulations. Lastly, the introduction of a co-solvent (Transcutol®) further enhanced the *in vitro* lipolysis performance for fenofibrate significantly. No significant loss in fenofibrate content was observed at long term and accelerated conditions over the course of three months for the optimized formulations.

CONCLUSION(S)

Following a systematic approach made the overall process of developing LBDDS for bioavailability enhancement more streamlined. Optimized combinations worked better than single excipient systems showing a synergistic effect for fenofibrate. Formulations with combinations of long chain and medium chain fatty acid esters were able to solubilize higher amount of fenofibrate during *in vitro* lipolysis compared to medium chain fatty acid esters on their own. At the same time medium chain fatty acid esters like Labrasol® ALF were needed to attain the desired drug load of 80mg/g. Lastly, no loss or degradation of fenofibrate was observed for SEDDS up to 3 months at long term and accelerated storage conditions.

REFERENCES

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